

Math 210A  
Final Exam  
December 10, 2021

Name:

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1. Is the group  $(\mathbb{Z}/16\mathbb{Z})^\times$  cyclic?

2. Prove that there is a group  $G$  that is not solvable but every proper subgroup of  $G$  is solvable.

3. Prove that if  $G$  has no non-trivial automorphisms, then  $g^2 = e$  for all  $g \in G$ .

4. Prove that every group of order 255 is cyclic.

5. Let  $G$  be a finite simple group and  $p$  a prime divisor of  $|G|$ . Prove that  $G$  is generated by elements of order  $p$ .

6. Prove that every group of order  $n$  is isomorphic to a subgroup of  $GL_n(\mathbb{Z})$ .

7. Let  $f$  and  $g$  be two morphisms between two objects  $X$  and  $Y$  in a category. Prove that the canonical morphism  $T \rightarrow \text{Coeq}(f, g)$  is an epimorphism.



8. Prove that every representable functor  $Sets \rightarrow Sets$  has a left adjoint.

9. Let  $f$  be a morphism in an abelian category such that  $\text{Ker}(f) = 0$  and  $\text{Coker}(f) = 0$ . Prove that  $f$  is an isomorphism. (You are not supposed to use Mitchell's theorem.)

10. Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be an additive functor of abelian categories. Prove that if  $F$  has a left adjoint functor, then  $F$  is left exact.